

Abnik Ahilasamy

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Education

Arizona State University — MS in Data Science, GPA 3.97/4.0

May 2026, Tempe, AZ

Coursework: Statistical Machine Learning, Data Processing at Scale, Computer Vision, Data Mining

SRM Institute of Science and Technology — B.Tech. in AI & ML (Top 5%), GPA 3.8/4

Jun 2024, Chennai, IN

Coursework: Deep Learning, Artificial Intelligence, Neural Networks, NLP, Nature-Inspired Computing, Algorithms

Experience

Larsen & Toubro — AI / LLM Engineer

Apr 2024 – Jun 2024, Chennai, IN

- Built **0 to 1 agentic RAG system** (LangChain, LlamaIndex), reducing hallucinations by **35%**
- Owned the **end-to-end pipeline** (retrieval, reasoning, APIs, evaluation, infra), delivering a production-ready system
- Re-architected inference and retrieval flow (caching, batching, memory optimization), achieving **40% lower latency** on workloads
- Designed **evaluation framework** (faithfulness, groundedness, nDCG@10, MRR) for rapid iteration
- Implemented **robust guardrails, failure handling** (PII filters, prompt constraints, rollback), reducing unsafe outputs by **50%**
- Deployed and scaled via **FastAPI + observability stack** (Prometheus, OpenTelemetry) with p95 latency SLOs, drift detection
- Rapidly iterated across prompts, retrieval strategies, and agent workflows, improving end-to-end answer quality under ambiguous requirements

Excelerate — Data Analyst

Aug 2023 – Sep 2023, Remote, USA

- Led distributed team of 4 across time zones, managing execution using structured workflows and data-driven prioritization
- Built BI dashboards (Power BI, Power Query, DAX) integrating data sources for real-time KPI tracking
- Performed data cleaning and analysis (Excel, formulas, pivot tables), improving decision efficiency by **30%**
- Delivered stakeholder reports, reducing analysis turnaround time by **25%** and improving decision clarity

LetsGrowMore (LGMVIP) — ML Engineer

Jun 2023 – Jul 2023, Chennai, IN

- Built end-to-end ML pipelines across NLP, time-series, and vision using Python and Scikit-learn
- Developed LSTM stock prediction and RNN text generation models, improving sequence prediction accuracy by **15–20%**
- Implemented CNN models for image classification and emotion detection, achieving **90%+ accuracy** on validation datasets
- Rapidly shipped 5+ ML projects, iterating across models and improving performance through tuning and evaluation

Projects

Vision-Language Model Fine-Tuning (Qwen2.5-VL-7B)

Feb 2026 – Mar 2026

- Fine-tuned VLM using **Unsloth + LoRA** ($r=16$, $\alpha=32$) for structured image-to-text generation with consistent schema outputs
- Identified dataset bias impacting generation; trained on **T4 (14.7GB VRAM)** with **7.6GB peak memory**, **5.5 min training**, validating efficient low-rank adaptation

Autonomous Research Agent with LLMs

Aug 2025 – Sep 2025

- Built full-stack AI research agent (FastAPI, Tailwind, Gemini), reducing summarization effort by **70%**
- Implemented observability and scalable agent framework (lmnr, Playwright), reducing failures **40%**, improving usability **95%**

Multi-Agent RAG with LLMs

Jul 2025 – Aug 2025

- Built multi-agent RAG system (5+ agents, tool-use), improving task efficiency **30%**
- Engineered LLM + retrieval pipelines (ReAct, memory), improving accuracy **25%**, reducing latency **40%**

Leadership & Activities

AI Technical Content Creator — LinkedIn

Apr 2025 – Present

- Built AI technical content presence, generating **50K+ impressions**, **26K+ reach**, and **600+ engagements** across posts.

Head of Events — WHHC & ACE Club, SRM — Event Lead

Aug 2022–Sep 2023

- Spearheaded hackathons and design challenges with **1,000+ participants**; ensured accessibility for all attendees.

Skills

Agentic AI & Orchestration: LangGraph, LangChain, CrewAI, AutoGen, MCP, Multi-Agent Systems, Tool Calling, Human-in-the-Loop, Knowledge Graphs, Vector Databases, Embeddings, FastAPI, Docker, AWS

RAG & Knowledge Systems: Advanced RAG, Agentic RAG, Hybrid Search, Query Routing, Reranking, LlamaIndex

Evaluation & Production AI: Context Engineering, AI Evals, LangSmith, Arize Phoenix, Observability, Guardrails, Monitoring

Certifications: DeepLearning.ai TensorFlow Specialization; Neural Networks & Deep Learning

Publications: Brain Tumor Classification with Transfer Learning (ResNet-50, IEEE 2023)